CTSE – Lab 06

Starting Kafka in the background

```
PS D:\YEAR4\Y4S2\Current Trends in Software Engineering - SE4010\Lab6> docker compose up -d
time="2026-03-10T11:44:03+05:30" level=warning msg="D:\\YEAR4\\Y4S2\\Current Trends in Software Engineering - SE4010\\Lab6\\docker-compose.yml: the attribute `version` is obsolete
, it will be ignored, please remove it to avoid potential confusion"
[+] up 7/12
 - Image confluentinc/cp-kafka:7.6.0 [⣿⣿⣿
```

Running Inventory Service

```
Run  InventoryServiceApplication  BillingServiceApplication  OrderServiceApplication  ApiGatewayApplication

 :: Spring Boot ::                (v4.0.3)

2026-03-10T11:55:29.039+05:30  INFO 228 --- [inventory-service] [  restartedMain] c.e.i.InventoryServiceApplication        : Starting InventoryServiceApplication using Java 21
2026-03-10T11:55:29.048+05:30  INFO 228 --- [inventory-service] [  restartedMain] c.e.i.InventoryServiceApplication        : No active profile set, falling back to 1 default p
2026-03-10T11:55:29.178+05:30  INFO 228 --- [inventory-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.dev
2026-03-10T11:55:29.178+05:30  INFO 228 --- [inventory-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider settin
2026-03-10T11:55:31.458+05:30  INFO 228 --- [inventory-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8082 (http)
2026-03-10T11:55:31.492+05:30  INFO 228 --- [inventory-service] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-10T11:55:31.492+05:30  INFO 228 --- [inventory-service] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.18]
2026-03-10T11:55:31.596+05:30  INFO 228 --- [inventory-service] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complet
2026-03-10T11:55:32.871+05:30  INFO 228 --- [inventory-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8082 (http) with context pa
2026-03-10T11:55:33.003+05:30  INFO 228 --- [inventory-service] [  restartedMain] o.a.kafka.common.config.AbstractConfig   : ConsumerConfig values:
    allow.auto.create.topics = true
    auto.commit.interval.ms = 5000
    auto.offset.reset = earliest
    bootstrap.servers = [localhost:9092]
    check.crcs = true
    client.dns.lookup = use_all_dns_ips
    client.id = consumer-inventory-group-1
    client.rack =
    connections.max.idle.ms = 540000
    default.api.timeout.ms = 60000
    enable.auto.commit = false
    enable.metrics.push = true
    exclude.internal.topics = true
    fetch.max.bytes = 52428800
    fetch.max.wait.ms = 500
    fetch.min.bytes = 1
```

Running Billing Service

```
Structure Alt+7  toryServiceApplication  BillingServiceApplication  OrderServiceApplication  ApiGatewayApplication

 :: Spring Boot ::                (v4.0.3)

2026-03-10T11:57:42.210+05:30  INFO 6712 --- [billing-service] [  restartedMain] c.e.b.BillingServiceApplication          : Starting BillingServiceApplication using Java 21.0.
2026-03-10T11:57:42.218+05:30  INFO 6712 --- [billing-service] [  restartedMain] c.e.b.BillingServiceApplication          : No active profile set, falling back to 1 default pr
2026-03-10T11:57:42.339+05:30  INFO 6712 --- [billing-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devt
2026-03-10T11:57:42.339+05:30  INFO 6712 --- [billing-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting
2026-03-10T11:57:44.428+05:30  INFO 6712 --- [billing-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8083 (http)
2026-03-10T11:57:44.459+05:30  INFO 6712 --- [billing-service] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-10T11:57:44.460+05:30  INFO 6712 --- [billing-service] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.18]
2026-03-10T11:57:44.556+05:30  INFO 6712 --- [billing-service] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization complete
2026-03-10T11:57:45.704+05:30  INFO 6712 --- [billing-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8083 (http) with context pat
2026-03-10T11:57:45.799+05:30  INFO 6712 --- [billing-service] [  restartedMain] o.a.kafka.common.config.AbstractConfig   : ConsumerConfig values:
    allow.auto.create.topics = true
    auto.commit.interval.ms = 5000
    auto.offset.reset = earliest
    bootstrap.servers = [localhost:9092]
    check.crcs = true
    client.dns.lookup = use_all_dns_ips
    client.id = consumer-billing-group-1
    client.rack =
    connections.max.idle.ms = 540000
    default.api.timeout.ms = 60000
    enable.auto.commit = false
    enable.metrics.push = true
    exclude.internal.topics = true
    fetch.max.bytes = 52428800
    fetch.max.wait.ms = 500
    fetch.min.bytes = 1
```

Running Order Service

```
Run   InventoryServiceApplication ×   BillingServiceApplication ×   OrderServiceApplication ×   ApiGatewayApplication ×

"C:\Program Files\Java\jdk-21\bin\java.exe" ...

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 :: Spring Boot ::                (v4.0.3)

2026-03-10T11:58:06.883+05:30  INFO 10504 --- [order-service] [  restartedMain] c.e.o.OrderServiceApplication            : Starting OrderServiceApplication using Java 21.0.5 w
2026-03-10T11:58:06.894+05:30  INFO 10504 --- [order-service] [  restartedMain] c.e.o.OrderServiceApplication            : No active profile set, falling back to 1 default pro
2026-03-10T11:58:07.049+05:30  INFO 10504 --- [order-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devto
2026-03-10T11:58:07.049+05:30  INFO 10504 --- [order-service] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting
2026-03-10T11:58:09.421+05:30  INFO 10504 --- [order-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8081 (http)
2026-03-10T11:58:09.459+05:30  INFO 10504 --- [order-service] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-10T11:58:09.460+05:30  INFO 10504 --- [order-service] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.18]
2026-03-10T11:58:09.563+05:30  INFO 10504 --- [order-service] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed
2026-03-10T11:58:10.667+05:30  INFO 10504 --- [order-service] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8081 (http) with context path
2026-03-10T11:58:10.683+05:30  INFO 10504 --- [order-service] [  restartedMain] c.e.o.OrderServiceApplication            : Started OrderServiceApplication in 4.817 seconds (pr
```

Running API Gateway

```
Run   InventoryServiceApplication ×   BillingServiceApplication ×   OrderServiceApplication ×   ApiGatewayApplication ×

"C:\Program Files\Java\jdk-21\bin\java.exe" ...

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 :: Spring Boot ::                (v4.0.3)

2026-03-10T11:58:48.868+05:30  INFO 15768 --- [api-gateway] [  restartedMain] c.e.api_gateway.ApiGatewayApplication    : Starting ApiGatewayApplication using Java 21.0.5 with
2026-03-10T11:58:48.880+05:30  INFO 15768 --- [api-gateway] [  restartedMain] c.e.api_gateway.ApiGatewayApplication    : No active profile set, falling back to 1 default profi
2026-03-10T11:58:49.068+05:30  INFO 15768 --- [api-gateway] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtool
2026-03-10T11:58:49.069+05:30  INFO 15768 --- [api-gateway] [  restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting th
2026-03-10T11:58:50.721+05:30  INFO 15768 --- [api-gateway] [  restartedMain] o.s.cloud.context.scope.GenericScope     : BeanFactory id=ee27788b-db68-349c-8f47-29c9978eba8c
2026-03-10T11:58:51.571+05:30  INFO 15768 --- [api-gateway] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat initialized with port 8084 (http)
2026-03-10T11:58:51.600+05:30  INFO 15768 --- [api-gateway] [  restartedMain] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-03-10T11:58:51.601+05:30  INFO 15768 --- [api-gateway] [  restartedMain] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/11.0.18]
2026-03-10T11:58:51.698+05:30  INFO 15768 --- [api-gateway] [  restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed i
2026-03-10T11:58:53.328+05:30  INFO 15768 --- [api-gateway] [  restartedMain] o.s.boot.tomcat.TomcatWebServer          : Tomcat started on port 8084 (http) with context path '
2026-03-10T11:58:53.357+05:30  INFO 15768 --- [api-gateway] [  restartedMain] c.e.api_gateway.ApiGatewayApplication    : Started ApiGatewayApplication in 5.647 seconds (proces
```
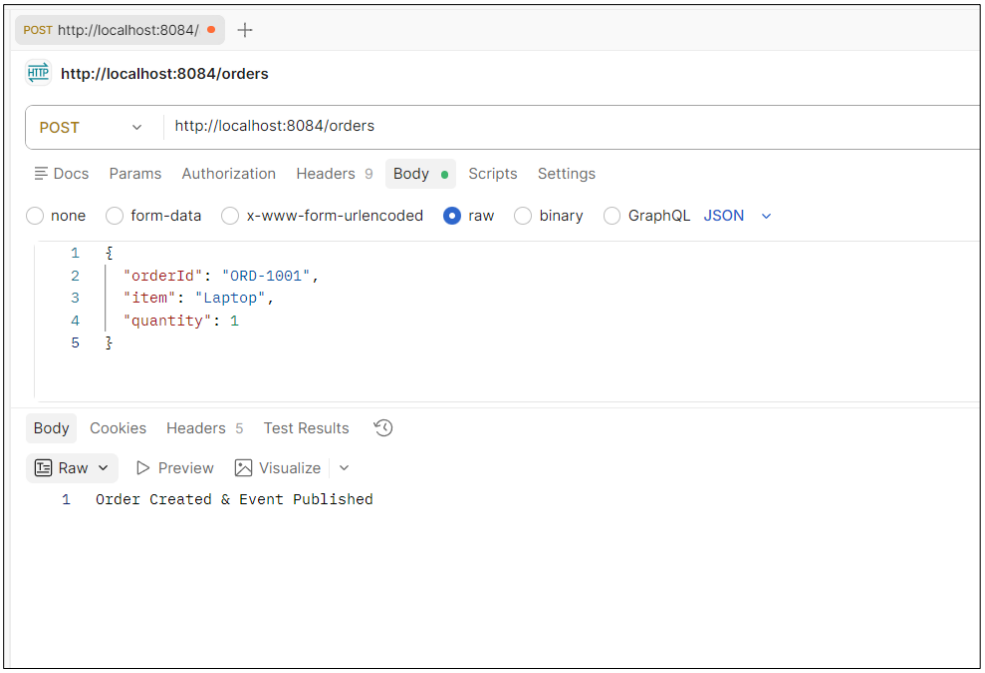
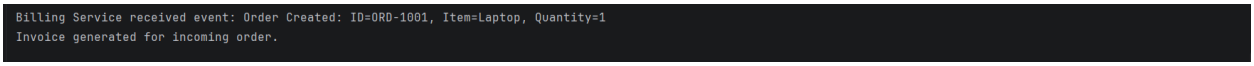

Test using Postman



In Order Service terminal

In Inventory Service terminal

In Billing Service terminal